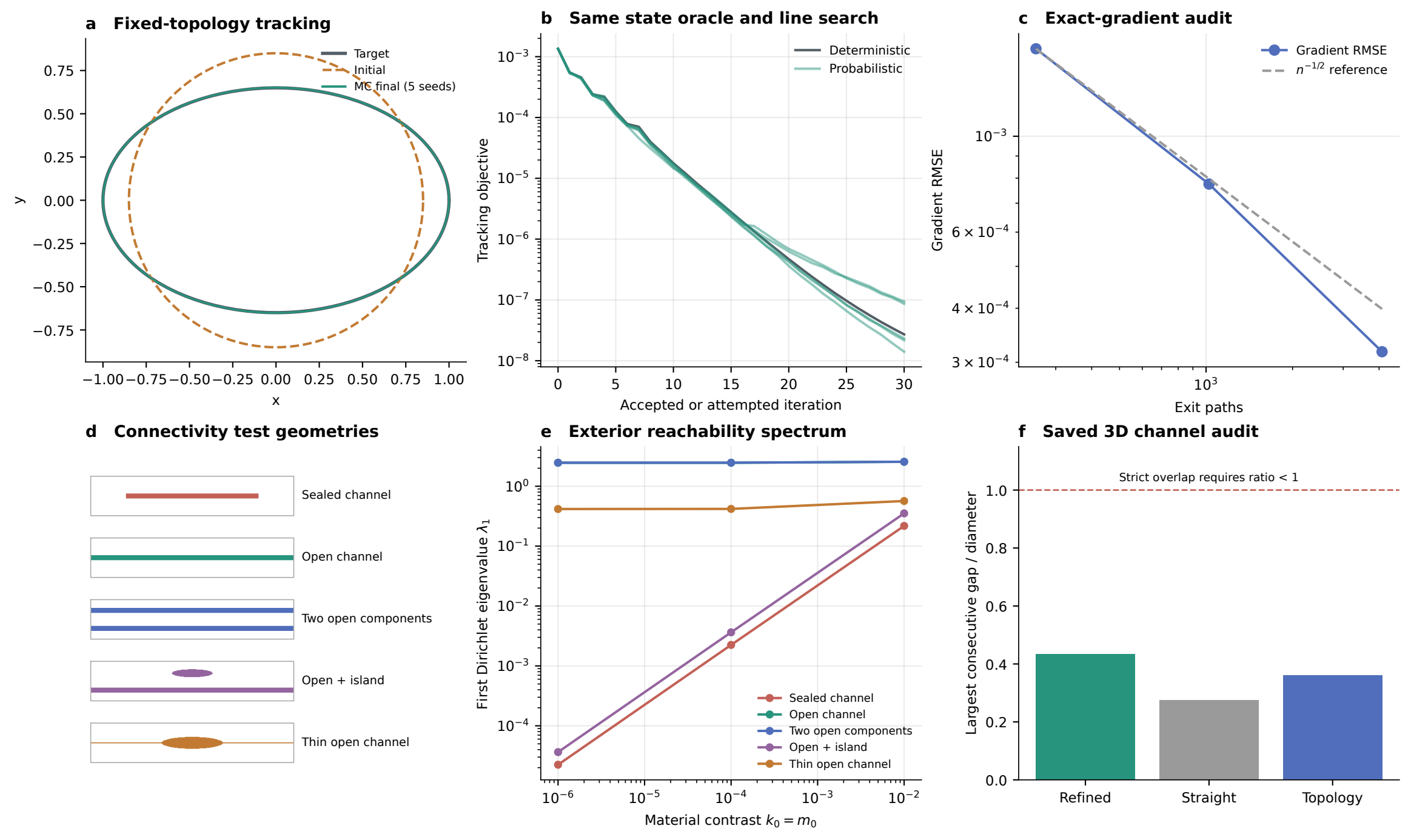

a Fixed-topology tracking
Target
Initial
MC final (5 seeds)
x
y
b Same state oracle and line search
Deterministic
Probabilistic
Tracking objective
Accepted or attempted iteration
c Exact-gradient audit
Gradient RMSE
$n^{-1/2}$ reference
Gradient RMSE
Exit paths
d Connectivity test geometries
Sealed channel
Open channel
Two open components
Open + island
Thin open channel
e Exterior reachability spectrum
First Dirichlet eigenvalue λ_1
Material contrast $k_0 = m_0$
f Saved 3D channel audit
Strict overlap requires ratio < 1
Largest consecutive gap / diameter
Refined
Straight
Topology